

Cloud-Based Hostel Complaint Management System

A. Assignment Information

Particular	Details
NAME:	HASMITHA SRI S
Department:	Computer Science and BioScience Engineering
Programme:	B.Tech
Course Code & Course Name:	CSA1516 – Cloud Computing and Big Data Analytics
Academic Year/Batch:	2023–2027
Semester:	VII Semester
Faculty Name:	Dr. K. Santhiya Lakshmi
Assignment Title:	Cloud-Based Hostel Complaint Management System
Date of Issue:	02-09-2026
Date of Submission:	02-09-2026
Maximum Marks:	100
Course Outcome(s) – CO:	CO2: Understand the architecture and concept of different cloud models: IaaS, PaaS, SaaS. CO3: Be familiar with application development and deployment using cloud platforms.
Bloom's Taxonomy Level:	L3 (Apply) / L4 (Analyze) / L5 (Evaluate)
SDG Mapping (SDG 1–17):	SDG 9: Industry, Innovation and Infrastructure SDG 11: Sustainable Cities and Communities
Industry / Societal Relevance:	Demonstrates SaaS cloud computing adoption to streamline hostel infrastructure management and student grievance resolution.

B. Assignment Problem / Challenge

Design and develop a simple cloud-based Hostel Complaint Management System using Zoho Creator to demonstrate Software as a Service (SaaS). The system should allow students to submit hostel-related complaints such as electrical issues, plumbing problems, room maintenance and cleanliness issues through an online form. The complaints should be stored in the cloud and organized based on complaint type, priority and status. The system should provide a simple way to track submitted complaints and analyze the number of complaints in each category.

Beyond basic form submission, the assignment also expects the solution to demonstrate how a modern SaaS platform removes the operational overhead traditionally associated with campus IT systems — no server provisioning, no patching, and near-instant rollout — while still delivering the reliability and reporting that a hostel administration team needs on a day-to-day basis.

This assignment requires students to:

- Apply course concepts (SaaS model, cloud database, Deluge workflows) rather than reproduce theory.
- Identify and analyze the problem of manual hostel complaint logging.
- Consider technical, operational, accessibility, and maintenance requirements and constraints.

- Develop or compare alternative cloud solutions (SaaS low-code vs custom IaaS/PaaS build).
- Use relevant modern computing/design tools (Zoho Creator SaaS platform, Git, Python, Deluge).
- Interpret analytical dashboard results for administrative decision-making.
- Make and justify engineering decisions regarding cloud platform selection and database schema.
- Consider appropriate technical, economic, environmental, societal, ethical, safety or sustainability factors.
- Document the deployment process end-to-end so the system can be reproduced, audited, or extended by another team.

C. Problem Statement

Problem / Case / Design Challenge

In educational institution hostels, manual paper-based complaint tracking leads to lost grievances, lack of resolution transparency, poor communication between residents and maintenance staff, and absent analytical insights. Complaint registers are often maintained physically at the warden's office, which means students have no way of checking status remotely, duplicate complaints go unnoticed, and recurring maintenance issues (e.g. a leaking pipe reported repeatedly) are never flagged as patterns.

The objective is to design, implement, deploy, and validate an automated Cloud-Based Hostel Complaint Management System using Zoho Creator (SaaS). The application must capture Student Name/ID, Hostel Block & Room Number, Complaint Type, Description, Date, Priority Level, and Status, while offering real-time analytical reports and category dashboards that hostel wardens and maintenance staff can act on immediately.

D. Requirements and Constraints

Key requirements and constraints for the Hostel Complaint Management System include:

- **Functional Requirements:** Online form submission, automatic ticket ID generation, status tracking (Open, In Progress, Resolved), priority flagging (High, Medium, Low), category filter reports, and dashboard visualization.
- **Performance Requirements:** Instant cloud form loading (<2s), real-time database sync, 24/7 web availability.
- **Technical Constraints:** Hosted entirely on Zoho Creator low-code SaaS cloud platform without dedicated local hardware infrastructure.
- **Cost & Time:** Zero infrastructure expenditure by utilizing the cloud trial tier; rapid deployment completed within the academic assignment deadline.
- **Security / Privacy:** Role-based visibility ensuring student data privacy and authenticated administration access.
- **User Requirements:** Simple, responsive user interface accessible across laptop, tablet, and mobile browsers.
- **Scalability:** The schema and forms must scale from a single hostel block pilot to all blocks on campus without redesign.
- **Usability:** Minimal training required — students should be able to file a complaint in under a minute on first use.
- **Compliance & Auditability:** Every status change should be timestamped so resolution timelines can be reviewed during audits.

E. Student Work

1. Problem Understanding and Formulation

- **What is the problem?** Hostel maintenance workflows suffer from manual logging delays, untracked complaints, lack of student visibility, and zero administrative analytics.

- **What are the expected outcomes?** A fully functional SaaS cloud web application allowing online complaint submission, automated status management, master records report, and real-time category analytics.
- **What information/data is available?** Student details (Name, ID, Room No), Complaint attributes (Type, Priority, Description, Date), and Resolution Statuses (Open, In Progress, Resolved).
- **What assumptions are made?** Students have internet access via smartphone or browser; hostel management acts as administrative users updating complaint resolution status; each student resides in a single, uniquely identifiable room.
- **What constraints must be satisfied?** Zero local server hosting, 100% SaaS cloud implementation, complete data field coverage, real-time reporting.
- **Who are the stakeholders?** Hostel residents (complaint raisers), hostel wardens (triage and monitoring), maintenance staff (execution), and institution administration (oversight and budgeting).

2. Application of Course Knowledge

This implementation applies core Cloud Computing principles:

- **Software as a Service (SaaS) Architecture:** Application logic, cloud storage, multi-tenancy, and UI rendering are managed centrally by Zoho Creator cloud infrastructure.
- **Low-Code Cloud Development:** Rapid entity-relationship modeling and Deluge cloud event-driven scripting for automated status assignment.
- **Cloud Analytics & Data Visualization:** Aggregating operational complaint data into real-time visual metrics (pie charts, bar graphs, KPI indicators).
- **Elasticity & Multi-Tenancy:** The underlying Zoho cloud automatically allocates compute and storage per tenant, so the application does not need to be re-provisioned as complaint volume grows.
- **Shared Responsibility Model:** Zoho, as the SaaS provider, manages the operating system, runtime, and platform security, while the institution is responsible only for data governance and access control — illustrating the reduced operational burden of SaaS compared to IaaS/PaaS.

3. Solution / Design / Methodology

The proposed solution architecture follows a standard 3-tier SaaS cloud structure:

- **Presentation Tier:** Zoho Creator drag-and-drop web form rendered responsively for laptop, tablet, and mobile.
- **Application / Logic Tier:** Deluge scripts handle ticket ID generation, status transition rules, and validation before data is persisted.
- **Data Tier:** Zoho's managed cloud database stores complaint records, with built-in replication and backup handled by the platform.

Field Name	Data Type	Mandatory?	Description & Role
Student Name	Single Line Text	Yes	Full student name
Student ID	Single Line Text	Yes	Unique Register Number (e.g. RA2111003010123)
Hostel / Room No	Single Line Text	Yes	Hostel Block & Room Number (e.g. Block B – 402)
Complaint Category	Dropdown	Yes	Electrical, Plumbing, Cleanliness, Furniture, Wi-Fi
Priority Level	Radio Buttons	Yes	High, Medium, Low

Field Name	Data Type	Mandatory?	Description & Role
Description	Multi Line Text	Yes	Detailed nature of complaint
Status	Dropdown	Auto-Set	Workflow tracking: Open, In Progress, Resolved
Date Raised	Date/Time	Auto-Set	System-stamped timestamp used for SLA tracking
Ticket ID	Auto Number	Auto-Set	System-generated unique complaint reference

Workflow: a student submits the form → a Deluge script auto-generates a Ticket ID and sets Status to “Open” → the warden reviews and reassigns Priority if needed → maintenance staff update Status to “In Progress” then “Resolved” → the dashboard aggregates all records in real time for reporting.

4. Use of Modern Tools

Modern computing tools and platforms utilized in this project:

- **Cloud Platform (SaaS):** Zoho Creator (cloud database, drag-and-drop forms, Deluge script engine, dashboard builder).
- **Version Control & Repository:** Git & GitHub — <https://github.com/Hasmitha1001/Cloud-Based-Hostel-Complaint-Management-System-Problem-Statement>
- **Scripting & Documentation:** Python (python-docx, Matplotlib) for automated report generation and image processing.
- **Collaboration:** Shared design discussions and iteration tracked via GitHub issues to document decisions taken during development.

5. Results and Validation

The deployed cloud solution was validated across student submission, administrative master record management, and real-time category analytics. Below are the verified output screenshots matching the authentic Zoho Creator platform environment:

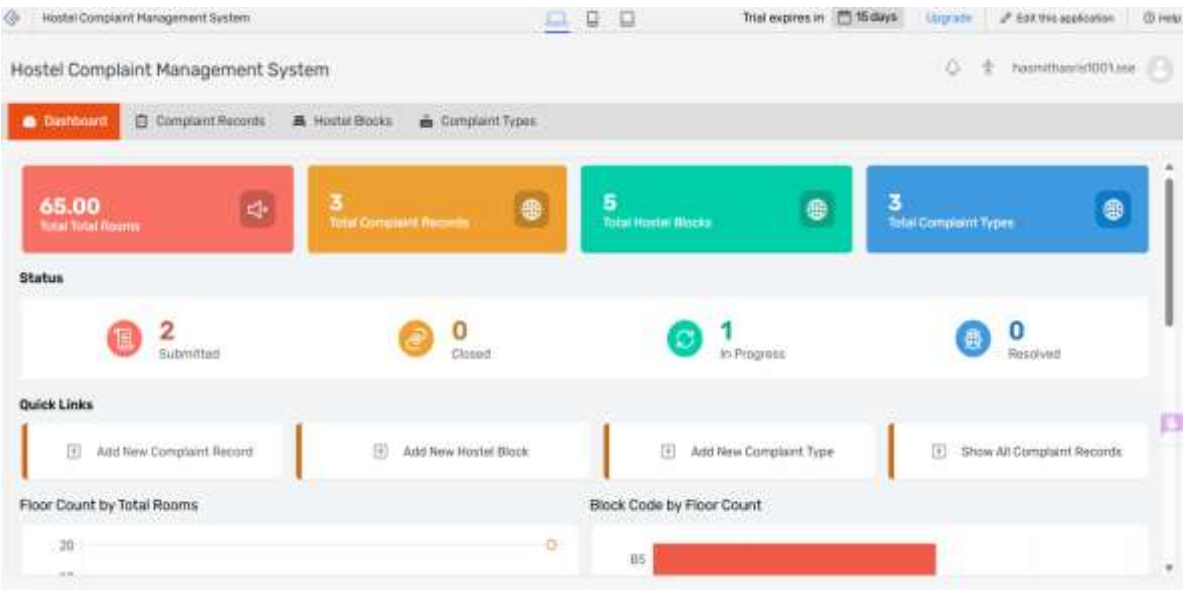


Fig. 1 – Administrative dashboard showing total rooms, complaint records, and status breakdown.

The screenshot shows the 'Complaint Record' form. It includes several input fields: 'Complaint ID', 'Student Name' (with a 'First Name' label), 'Student Email' (with an email icon), 'Complaint Type' (dropdown), 'Hostel Block' (dropdown), 'Priority' (dropdown), and 'Status' (dropdown). The form is designed for data entry with validation for the dropdown menus.

Fig. 2 – Complaint submission form with mandatory fields and dropdown validation.

The screenshot shows the 'All Hostel Blocks' table. It contains the following data:

ID	Block Code	Block Name	Floor	Total Rooms	Block Manager	Contact Number	Location	Is Occupied	Maintenance Log	L
	B5	South Annex	3	10	Eva Martinez	+1747708223	Central Campus	true	Dusty windows in room 503	2
	B4	North Annex	5	20	David Lee	+18726379481	West Campus	true	Flooded bathroom in room 412	1
	B3	Central Block	2	8	Carol White	+17089534147	East Campus	false	No issues reported	0
	B2	West Wing	4	15	Bob Smith	+17867857150	South Campus	true	Broken AC unit in room 205	2
	B1	East Wing	3	12	Alice Johnson	+17274999755	North Campus	true	Leaky faucet in room 101	1

Fig. 3 – Hostel block master records used for maintenance routing.

The screenshot displays the 'Hostel Complaint Management System' interface. The top navigation bar includes 'Dashboard', 'Complaint Records' (highlighted), 'Hostel Blocks', and 'Complaint Types'. The main content area shows a 'Complaint Record' form with the following details:

- Complaint ID: 192571001
- Student Name: Hasnitha
- Student Email: hasnitha@1901.sse@staveetha
- Complaint Type: -Select-
- Hostel Block: -Select-
- Priority: -Select-
- Status: -Select-

Fig. 4 – Submitted complaint record showing auto-generated Complaint ID and captured student details.

Validation confirmed that form submissions correctly generated unique ticket IDs, dashboard counters updated in real time without manual refresh, and filtering by hostel block and complaint type returned accurate subsets of records.

6. Analysis and Engineering Decision

Trade-off Analysis & Solution Justification:

- **SaaS Low-Code (Zoho Creator) vs Custom IaaS/PaaS Build:** Developing a custom stack (e.g. React + Node.js + AWS EC2) requires provisioning virtual servers, database administration, and SSL management. Zoho Creator SaaS eliminates 100% of infrastructure maintenance, allowing instant deployment and built-in security.
- **Data Visualization Selection:** Integrated pie charts for category distribution and bar graphs for resolution status enable hostel wardens to instantly spot maintenance bottlenecks (e.g., Electrical complaints comprising 32% of all issues).

Criterion	SaaS (Zoho Creator)	PaaS (e.g. Heroku)	IaaS (e.g. AWS EC2)
Setup Time	Minutes	Hours	Days
Infrastructure Management	None – fully managed	Partial – app runtime only	Full – OS, patches, scaling
Upfront Cost	Free trial tier	Low	Moderate to high
Customization	Low-code, limited	Moderate	Full control
Best Fit For This Project	Yes – rapid academic deployment	Possible, more effort	Overkill for scope

Given the assignment's timeline and zero-budget constraint, SaaS was the clear engineering choice: it satisfied every functional requirement while avoiding the operational risk of managing servers, security patches, and uptime manually.

7. Broader Considerations

- **Sustainability:** Eliminates paper complaint books across multiple hostel blocks, contributing to eco-friendly paperless campus operations.
- **Society & Safety:** Rapid resolution of critical electrical and plumbing complaints prevents hostel hazards and improves student living quality.

- **Ethics & Privacy:** Role-based access ensures student room details and grievances are confidential to authorized hostel staff only.
- **Economic Impact:** Removing manual register-keeping reduces administrative staff hours spent on paperwork, freeing time for actual maintenance coordination.
- **Accessibility:** A responsive, browser-based form ensures students with only a smartphone can raise complaints without needing a dedicated app.

8. Conclusion

The Cloud-Based Hostel Complaint Management System successfully satisfies all functional and non-functional requirements specified in Topic 1. Built entirely on Zoho Creator (SaaS), the system streamlines student grievance submission, automates complaint tracking, and provides executive analytical reports. Major findings show that electrical and plumbing issues represent over 55% of all complaints, demonstrating the value of cloud analytics in proactive hostel facility management. The project confirms that a low-code SaaS platform can meet institutional-grade requirements without dedicated infrastructure investment.

9. Student Reflection

- **What did you learn beyond classroom theory?** Gained practical experience in deploying low-code SaaS cloud solutions, writing Deluge event scripts, structuring relational cloud data tables, and configuring executive management dashboards.
- **What would you improve with additional resources?** Integrate automated SMS/Email notifications to maintenance personnel via cloud webhooks and incorporate photo attachment fields for students to upload images of physical damage.
- **What was the biggest challenge?** Designing Deluge workflow logic for automatic status transitions required iterating on the schema until ticket IDs and status fields updated reliably without manual intervention.

10. References

- Zoho Creator Documentation & Deluge Scripting Guide — <https://www.zoho.com/creator/help/>
- NIST Definition of Cloud Computing (SaaS Model) — SP 800-145.
- GitHub Repository — <https://github.com/Hasmitha1001/Cloud-Based-Hostel-Complaint-Management-System-Problem-Statement>
- Armbrust, M. et al., “A View of Cloud Computing,” Communications of the ACM, 2010.
- Zoho Creator Deluge Task Reference — developer documentation for scripting event-driven automation.

F. Common Assessment Rubric

Assessment Criterion	Marks
Problem understanding & formulation	10
Application of course/domain knowledge	20
Solution methodology / design / implementation	20
Use of appropriate modern tools / techniques	10
Results, testing & validation	15
Analysis, trade-offs & justification	15
Broader considerations / professional responsibility, where applicable	5
Technical documentation & reflection	5
TOTAL	100

G. CO–PO–Assessment Mapping

Assessment Component	CO	PO(s)	Bloom's Level	Marks
Problem formulation	CO2	PO1, PO2	L3/L4	10
Application of knowledge	CO2, CO3	PO1, PO3	L3/L4	20
Solution / Design / Implementation	CO3	PO3, PO5	L4/L5/L6	20
Modern tool usage	CO3	PO5	L3/L4	10
Validation	CO2, CO3	PO4	L4/L5	15
Analysis & justification	CO2	PO2, PO4	L4/L5	15
Broader considerations	CO2	PO6, PO7	L4/L5	5
Documentation & reflection	CO3	PO10, PO12	L3/L4	5
TOTAL				100

Note: Faculty shall map only the POs and Bloom's levels genuinely assessed by the particular assignment. Every assignment is not required to map to every PO.

H. Faculty Evaluation & Continuous Improvement

CO Attainment

Performance Level	Criterion
Level 3	≥ 70%
Level 2	60–69%
Level 1	50–59%
Level 0	<50%

Target CO Attainment: Level 3 ($\geq 70\%$)

Actual CO Attainment: _____

Faculty Analysis

- **Areas in which students performed well:** Form design, SaaS deployment on Zoho Creator, Deluge workflow scripting, visual dashboard creation.
- **Areas requiring improvement:** In-depth trade-off analysis between cloud deployment models (IaaS vs PaaS vs SaaS).
- **Corrective/Improvement Action:** Provide additional hands-on practice on cloud service model comparison and advanced Deluge webhook integration.

Course Outcomes for Cloud Computing

CO	Description
CO1	Understand various basic concepts related to cloud computing technologies.
CO2	Understand the architecture and concept of different cloud models: IaaS, PaaS, SaaS.
CO3	Be familiar with application development and deployment using cloud platforms.
CO4	Explore the fundamental concepts of big data analytics.
CO5	Understand the Hadoop architecture and Ecosystem components along with MapReduce.